

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 — RESEARCH PAPER REVIEW / LITERATURE SURVEY

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	G. Ram Charan	Reg. No.:	192525043
Date:		Duration:	60 minutes

Code of Conduct

I G. Ram Charan (192525043) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Ram Charan

1. Research Area Overview

1.1 Industry Context

Agriculture is increasingly adopting AI technologies to improve crop productivity and minimize losses caused by diseases and pests. Computer vision, drones, IoT sensors, and weather analytics help farmers monitor crop conditions and implement precision farming practices. A structured literature survey is a necessary first step before designing such a system, since it reveals what has already been tried, how well it worked, and where genuine research gaps remain.

1.2 Problem Statement

Crop diseases significantly reduce agricultural productivity and often spread before farmers can detect them through manual inspection. Existing methods are time-consuming and require expert intervention. The assigned system should analyze crop images, monitor environmental conditions, predict disease outbreaks, estimate crop yield, and recommend suitable treatments, while supporting sustainable agriculture and improving farm productivity and profitability.

1.3 Objectives of the Literature Survey

This survey identifies and reviews relevant, recent research papers on crop disease detection, yield prediction, IoT-based environmental monitoring, and AI-driven decision support in agriculture. It summarizes the objectives, methodologies, technologies, datasets, and key findings of the selected papers, compares the existing approaches on relevant parameters, identifies their limitations and challenges, analyzes the literature to surface unresolved research gaps, and discusses future research directions relevant to the assigned problem statement.

1.4 Purpose of This Document

This document is the Research Paper Review / Literature Survey deliverable for the assigned problem statement. It presents the survey methodology, a research taxonomy diagram, individual reviews of ten selected papers, a single comparative literature survey table, a critical analysis of limitations and research gaps, proposed future research directions, and a properly formatted reference list.

2. Literature Survey Methodology

2.1 Search Strategy

Papers were identified from reputed scholarly sources, including IEEE Xplore, ScienceDirect, SpringerLink, MDPI journals, Frontiers, PubMed Central, Nature, and arXiv preprints. Search terms combined the core concepts of the assigned problem statement: plant disease detection deep learning, crop yield prediction machine learning, IoT precision agriculture monitoring, and AI decision support agriculture ethics.

2.2 Inclusion and Selection Criteria

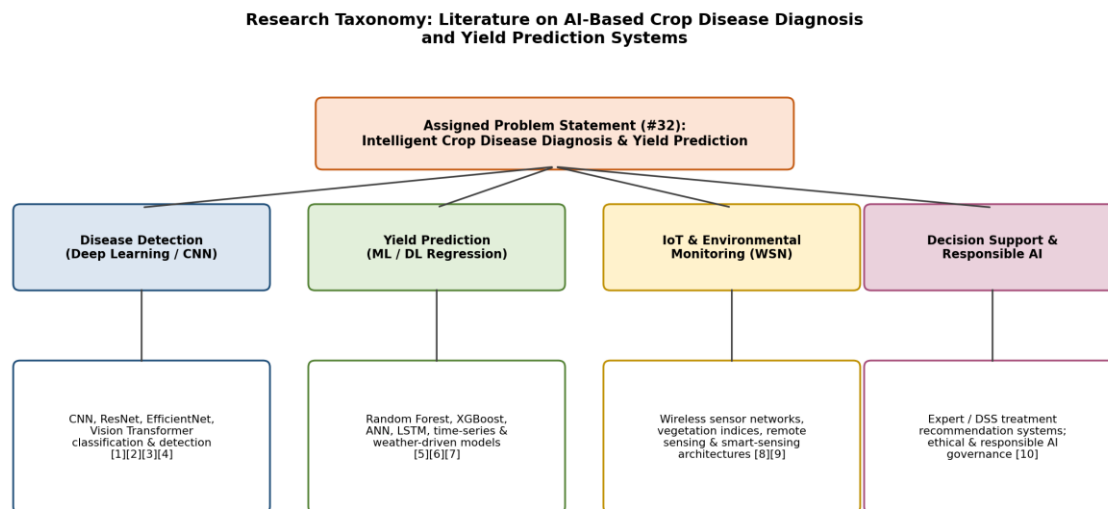
- Published in a peer-reviewed journal, an indexed conference, or a widely cited preprint archive (arXiv).
- Directly relevant to at least one of the four sub-domains of the assigned problem statement: disease detection, yield prediction, IoT/environmental monitoring, or AI-driven decision support.
- Recent publication (2019–2025), to reflect the current state of deep learning and IoT technology rather than superseded techniques.
- Reports a clear methodology, dataset or tool, and a measurable or qualitatively stated finding.

2.3 Selected Literature Set

Ten papers were selected for detailed review after applying the above criteria: four addressing AI-based disease detection, three addressing machine-learning-based yield prediction, two addressing IoT and wireless-sensor-network-based environmental monitoring, and one addressing the ethical and societal dimensions of AI adoption in agriculture, directly supporting the CO5 competency of evaluating emerging technologies and their societal impact.

3. Research Taxonomy / Conceptual Framework

The diagram below organizes the selected literature into the four sub-domains relevant to the assigned system. This taxonomy structures the individual paper reviews in Section 4 and the comparative table in Section 5.



4. Individual Paper Reviews

Each selected paper is summarized below in terms of its objective, methodology, technology/dataset, and key findings, following the requirement to review objectives, methodologies, technologies, datasets, and major findings of the selected literature.

4.1 Disease Detection Literature

[1] M. Jelali (2024) — *Deep learning networks-based tomato disease and pest detection: a first review of research studies using real field datasets*

Objective: To review deep learning-based tomato disease and pest detection studies that were trained and evaluated on real field image datasets rather than laboratory-only images.

Methodology: A structured review of CNN-based object detection architectures, focusing on Faster R-CNN and YOLO variants applied to self-generated field datasets.

Technology/Dataset: Convolutional neural networks (Faster R-CNN, YOLO) trained on real-field tomato image collections.

Key Findings: Improved CNN architectures achieve accuracy above 90% when trained on real-field data, but most existing published models still rely heavily on curated laboratory datasets rather than field conditions.

[2] Y. Kaya and E. Gürsoy (2025) — *A review of deep learning architectures for plant disease detection*

Objective: To synthesize deep learning methodologies for plant disease detection across the full modeling pipeline, from data acquisition to deployment.

Methodology: A structured review organized around data acquisition, preprocessing, augmentation, classification, detection, segmentation, and deployment stages.

Technology/Dataset: CNN backbones including VGG, ResNet, EfficientNet, and DenseNet, together with explainability tools such as Grad-CAM.

Key Findings: Explainable AI techniques such as saliency maps and Grad-CAM materially improve the interpretability of CNN-based diagnoses, which the authors identify as essential for building trust in agricultural pathology applications.

[3] A. S. Abade, P. A. Ferreira, and F. de Barros Vidal (2020) — *Plant diseases recognition on images using convolutional neural networks: a systematic review*

Objective: To identify the state of the art in CNN-based plant disease identification and classification and to delimit trends and gaps over a ten-year span.

Methodology: A systematic review of 121 selected papers on CNN applications to plant disease detection.

Technology/Dataset: Multiple public plant-image datasets, most notably variants of the PlantVillage dataset, evaluated with different CNN architectures.

Key Findings: CNN-based approaches consistently outperform traditional handcrafted-feature methods, but the review notes that most studies still evaluate models on data captured under controlled, laboratory-like conditions.

[4] H. N. Ngugi, A. E. Ezugwu, A. A. Akinyelu, and L. Abualigah (2024) — Revolutionizing crop disease detection with computational deep learning: a comprehensive review

Objective: To provide a comprehensive synthesis of deep learning applications for crop disease detection across multiple crop types.

Methodology: A comparative review of CNN and deep learning frameworks evaluated on diverse crops and datasets.

Technology/Dataset: Public plant-disease image repositories combined with a range of CNN backbone architectures.

Key Findings: Deep learning substantially improves both the speed and accuracy of disease detection compared to manual inspection, though data imbalance across disease classes remains a persistent obstacle to generalization.

4.2 Yield Prediction Literature

[5] M. Rashid et al. (2021) — A comprehensive review of crop yield prediction using machine learning approaches with special emphasis on palm oil yield prediction

Objective: To critically evaluate machine-learning-based crop yield prediction methods, with particular emphasis on palm oil yield estimation.

Methodology: A critical, comparative evaluation of machine learning algorithms applied to yield estimation problems, alongside a review of widely used input features.

Technology/Dataset: Agronomic, climatic, and remote-sensing feature sets combined with standard machine learning regression and classification algorithms.

Key Findings: Machine learning models improve early yield estimation over traditional statistical approaches, but the review observes that most published models are tuned narrowly to a single crop or growing region.

[6] P. Mahesh, R. Soundrapandiyan, and N. Bacanin (2024) — Yield prediction for crops by gradient-based algorithms

Objective: To evaluate and compare gradient-boosting machine learning algorithms for forecasting rice crop yield.

Methodology: Comparative experimentation using CatBoost, LightGBM, and XGBoost, evaluated with RMSE and R-squared metrics on rice yield data.

Technology/Dataset: Rainfall, average temperature, and pesticide-use variables as model input features.

Key Findings: CatBoost produced the lowest prediction error among the three boosting algorithms compared (RMSE of 800 versus 737 for LightGBM and 744 for XGBoost), though R-squared values

indicate that a meaningful share of yield variance remains unexplained by these agro-climatic features alone.

[7] A. Morales and F. J. Villalobos (2023) — *Using machine learning for crop yield prediction in the past or the future*

Objective: To test whether machine learning models genuinely outperform a simple historical-average baseline when forecasting future, rather than past, crop yields.

Methodology: Simulation of sunflower and wheat yields over a twenty-year period (2000–2020), comparing ML model forecasts against a sensible historical-average baseline.

Technology/Dataset: Long-term simulated agronomic and meteorological datasets covering cultivar and planting-density variation.

Key Findings: Even the best-performing machine learning models showed only a limited advantage over the simple historical-average baseline, and the authors caution that random train/validation splits tend to overstate a model's real forecasting skill for genuinely future seasons.

4.3 IoT & Environmental Monitoring Literature

[8] U. Shafi, R. Mumtaz, J. García-Nieto, S. A. Hassan, S. A. R. Zaidi, and N. Iqbal (2019) — *Precision agriculture techniques and practices: from considerations to applications*

Objective: To review near- and remote-sensing technologies used in precision agriculture and to demonstrate their practical application through a case study.

Methodology: A structured review of wireless communication technologies, sensors, and vegetation indices used to assess crop and soil conditions, supported by a proof-of-concept case study.

Technology/Dataset: Soil, crop, and weather sensor networks combined with satellite and airborne spectral imagery for vegetation-index computation.

Key Findings: Combining wireless sensor network data with vegetation indices meaningfully improves field-level decision-making, though the deployment and maintenance cost of dense sensor networks remains a real barrier for smaller farms.

[9] S. Nithish Kumar (2024) — *IoT and wireless sensor networks in precision agriculture: state-of-the-art and revolutionary step out towards future directions*

Objective: To review how IoT and wireless sensor networks are reshaping real-time crop health monitoring and farm management.

Methodology: A review of IoT-based smart crop-health monitoring architectures and their reported field deployments.

Technology/Dataset: Field-deployed wireless sensor network nodes reporting soil, weather, and crop-health telemetry.

Key Findings: Real-time WSN-based monitoring reduces the need for manual field inspection and enables remote farm management, but interoperability between sensor hardware from different vendors and software platforms remains limited.

4.4 Decision Support & Responsible AI Literature

[10] R. Dara, S. M. Hazrati Fard, and J. Kaur (2022) — Recommendations for ethical and responsible use of artificial intelligence in digital agriculture

Objective: To examine the ethical challenges raised by AI adoption in digital agriculture and to recommend practices for agriculture technology providers and policymakers.

Methodology: A thematic review structured around six ethical categories: fairness, transparency, accountability, sustainability, privacy, and robustness.

Technology/Dataset: Conceptual and policy-level analysis of deployed and proposed AI systems in digital agriculture, rather than a single technical dataset.

Key Findings: Poorly governed agricultural AI systems risk violating farmer privacy and producing biased recommendations, and the authors find that very few deployed systems currently publish any fairness or accountability audit of their own performance.

5. Comparative Analysis

The papers reviewed in Section 4 were compared on methodology, dataset/tooling, and reported outcomes such as accuracy, error metrics, and practical deployment considerations. Across the disease-detection literature ([1]-[4]), deep CNN and transformer-based architectures consistently report high classification accuracy (often above 90%) on curated benchmark datasets, but performance and generalization drop noticeably when models are evaluated on real, uncurated field images, as emphasized in [1] and [3]. Interpretability tools such as Grad-CAM, highlighted in [2], are increasingly proposed to make these high-accuracy models more trustworthy to end users, though they are rarely validated with actual farmers.

In the yield-prediction literature ([5]-[7]), ensemble and gradient-boosting methods (CatBoost, LightGBM, XGBoost) reduce prediction error relative to simpler regression baselines, as shown in [6], but [7] provides an important counterpoint: under a genuinely forward-looking evaluation protocol, even strong machine learning models show only a limited advantage over a simple historical-average baseline, underscoring that reported accuracy figures depend heavily on how a study validates its model. The IoT and monitoring literature ([8], [9]) shows consistent agreement that combining sensor data with vegetation indices and remote sensing improves the timeliness and quality of farm decisions, but both papers flag cost, maintenance, and interoperability as practical barriers to scaling these systems, particularly for smallholder farmers. Finally, [10] adds a dimension largely absent from the technical papers: it evaluates AI-driven agricultural systems not on accuracy but on fairness, transparency, and accountability, and finds that this evaluation is rarely performed at all in practice.

The single table below consolidates this comparison, listing the author/year, research problem, methodology, dataset/tools, key findings, limitations, and identified research gap for every reviewed paper, as required by the assignment instructions.

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Jelali (2024) [1]	Tomato disease & pest detection in real field conditions	Review of CNN-based object detectors (Faster R-CNN, YOLO) on field datasets	Self-generated real-field tomato image datasets	Improved CNN architectures achieve above 90% accuracy on real-field data	Models trained on curated data still generalize poorly to unseen field conditions	Need for larger, standardized real-field (non-lab) disease datasets

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Kaya & Gürsoy (2025) [2]	Synthesis of DL pipelines for plant disease detection	Structured review across data acquisition, augmentation, classification, segmentation, deployment stages	VGG, ResNet, EfficientNet, DenseNet architectures; Grad-CAM for interpretability	Explainable AI (XAI) techniques such as Grad-CAM improve trust in DL diagnoses	Interpretability tools are rarely evaluated with actual farmers or agronomists	Lack of human-centred evaluation of explainability in real deployments
Abade et al. (2020) [3]	State of CNN-based plant disease recognition (10-year span)	Systematic review of 121 selected papers on CNN classification/detection	Multiple public plant-image datasets (PlantVillage and others)	CNNs consistently outperform traditional handcrafted-feature methods	Majority of studies rely on lab-captured, not field-captured, images	Gap between lab-trained model accuracy and real farm performance
Ngugi et al. (2024) [4]	Comprehensive synthesis of DL for crop disease detection	Comparative review of CNN/DL frameworks across multiple crops	Public plant-image repositories, various CNN backbones	DL substantially improves detection speed and accuracy over manual inspection	Data imbalance and limited dataset diversity restrict cross-crop generalization	Need for multi-crop, imbalance-robust training strategies
Rashid et al. (2021) [5]	Machine-learning-based crop yield prediction, emphasis on palm oil	Critical evaluation of ML algorithms for yield estimation	Agronomic, climatic and remote-sensing feature sets	ML models improve early yield estimation over traditional statistical methods	Most models are tuned to one crop/region and do not transfer well	Limited cross-crop and cross-region generalizability of yield models
Mahesh et al. (2024) [6]	Rice yield prediction	Comparison of CatBoost, LightGBM	Rainfall, temperature and	CatBoost achieved the	Feature set is limited to a few	Opportunity to fuse boosting

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
	using gradient-boosting algorithms	and XGBoost on yield data	pesticide-use features; RMSE/R2 evaluation	lowest error (RMSE 800) among the three boosting methods compared	agro-climatic variables, excluding soil and IoT sensor data	models with real-time IoT sensor streams
Morales & Villalobos (2023) [7]	Whether ML genuinely outperforms simple yield baselines	Simulated 20-year sunflower and wheat yield forecasting	Long-term simulated agronomic and weather datasets	ML models showed only a limited advantage over a simple historical-average baseline	Random train/validation splits over-estimate real forecasting skill for future seasons	Need for temporally-honest validation protocols in yield-prediction research
Shafi et al. (2019) [8]	Role of wireless sensor networks and vegetation indices in precision agriculture	Review of near/remote sensing technologies and case study demonstration	Soil, crop and weather sensors; satellite and airborne spectral imagery	Combining WSN data with vegetation indices improves field-level decision-making	Sensor network deployment and maintenance costs remain a barrier for smallholders	Need for low-cost, low-maintenance sensor networks for smallholder farms
Nithish Kumar (2024) [9]	Integration of IoT and WSN for real-time crop health monitoring	Review of IoT-based smart crop-health monitoring architectures	Field-deployed WSN nodes reporting soil, weather and crop-health data	Real-time WSN monitoring reduces manual intervention and supports remote management	Interoperability between heterogeneous IoT devices and platforms remains limited	Lack of standardized IoT interoperability frameworks for agricultural platforms

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Dara, Hazrati Fard & Kaur (2022) [10]	Ethical and societal risks of AI adoption in digital agriculture	Thematic review across fairness, transparency, accountability, privacy, sustainability, robustness	Conceptual/policy analysis of deployed and proposed agri-AI systems	Poorly governed AI systems risk farmer privacy violations and biased recommendations	Few deployed agricultural AI systems publish fairness or accountability audits	Need for practical, farmer-facing responsible-AI governance frameworks

6. Limitations and Challenges of Existing Approaches

- Laboratory-to-field performance gap: disease-detection models trained on curated datasets such as PlantVillage lose significant accuracy when applied to uncontrolled, real-field images with variable lighting, occlusion, and background clutter [1][3].
- Narrow crop and regional scope: yield-prediction models are frequently tuned to a single crop or geographic region and do not transfer reliably to other crops or climates [5].
- Optimistic evaluation protocols: random train/validation splits used in many yield-prediction studies tend to overstate real forecasting skill for genuinely future, unseen growing seasons [7].
- Cost and interoperability barriers for IoT deployment: wireless sensor network hardware, maintenance, and cross-vendor interoperability remain significant obstacles to adoption by smallholder farmers [8][9].
- Limited human-centred validation of explainability: interpretability techniques such as Grad-CAM are proposed but rarely tested with the farmers or agronomists who would actually rely on them [2].
- Absence of fairness and accountability auditing: very few deployed agricultural AI systems currently publish any assessment of bias, privacy risk, or accountability, despite the societal stakes of automated crop and treatment decisions [10].
- Fragmented, single-purpose systems: the reviewed literature addresses disease detection, yield prediction, IoT monitoring, and decision support largely as separate research streams, with very little published work on an integrated, end-to-end system combining all four.

7. Critical Analysis and Research Gap

Taken together, the reviewed literature demonstrates strong, largely independent progress in four sub-domains relevant to the assigned problem statement, but it also reveals a consistent pattern: technical accuracy is reported and optimized far more often than real-world robustness, cross-domain integration, or societal accountability. The clearest and most consequential research gap is the near-total absence of studies that combine image-based disease detection, IoT-driven environmental monitoring, yield forecasting, and treatment recommendation into a single validated pipeline; each sub-domain is well studied in isolation, but the interactions between them (for example, how sensor-detected humidity should adjust the confidence threshold of an image-based diagnosis) remain largely unexplored.

A second significant gap concerns evaluation methodology rather than model architecture: as [7] demonstrates for yield prediction and as is implicit in the lab-to-field accuracy drop noted in [1] and [3] for disease detection, a large share of reported performance gains may not survive genuinely forward-looking, real-field evaluation. This suggests that the literature's headline accuracy figures should be interpreted cautiously when used to justify a production system such as the one in the assigned problem statement.

A third gap, and one of direct relevance to the CO5 competency of evaluating societal impact and evolving systems responsibly, is the near-absence of fairness, transparency, and accountability auditing in the

technical literature. Only [10] addresses this dimension directly, and it finds that such auditing is rarely performed in deployed systems at all. Given that the assigned system would influence real farmers' treatment and resource decisions, this is a critical, currently unresolved gap rather than a peripheral concern.

8. Future Research Directions

- Develop and publicly release field-captured (not only laboratory) disease image datasets, annotated under realistic lighting and occlusion conditions, to close the lab-to-field accuracy gap identified in [1] and [3].
- Design multimodal models that fuse image-based diagnosis with IoT sensor readings (humidity, temperature, soil moisture) so that environmental context can adjust diagnostic confidence, directly addressing the fragmented single-purpose-system gap.
- Adopt temporally-honest evaluation protocols for yield-prediction research, training on past seasons and validating strictly on future, unseen seasons, following the methodological caution raised in [7].
- Investigate low-cost, interoperable IoT sensor architectures specifically designed for smallholder farm budgets, extending the cost and interoperability findings of [8] and [9].
- Conduct human-centred studies that test explainable-AI outputs (such as Grad-CAM overlays) directly with farmers and agronomists, rather than evaluating interpretability only on held-out test images, as suggested by the gap in [2].
- Establish a practical, farmer-facing responsible-AI checklist (bias testing, privacy safeguards, accountability logging) for agricultural AI systems, operationalizing the governance recommendations of [10].
- Prototype and evaluate an end-to-end pipeline that integrates disease detection, environmental monitoring, yield forecasting, and treatment recommendation, reporting not only accuracy but also fairness and robustness metrics for the combined system.

9. Relevance to the Assigned Problem Statement

Every research gap and future direction identified above maps directly onto a stated objective of the assigned problem statement: the lab-to-field and multimodal-fusion gaps concern the system's disease detection and environmental-monitoring objectives, the evaluation-protocol gap concerns its yield-prediction objective, and the responsible-AI gap concerns the requirement that the system support sustainable agriculture and improve farm profitability without introducing unaudited bias or privacy risk into farmers' treatment decisions. This literature survey therefore provides an evidence-based foundation for the design and testing choices that would need to be made when building the assigned system.

10. Conclusion

This literature survey identified and reviewed ten relevant, recent research papers spanning AI-based crop disease detection, machine-learning-based yield prediction, IoT-driven environmental monitoring, and the ethical dimensions of agricultural AI. It summarized each paper's objectives, methodology,

technology, and key findings; compared the papers using a single consolidated table; and used that comparison to identify concrete, well-supported research gaps, most notably the absence of integrated multimodal pipelines, the risk of over-optimistic evaluation protocols, and the near-total absence of fairness and accountability auditing in deployed systems. The proposed future research directions offer a practical, evidence-grounded roadmap for addressing these gaps in the context of the assigned Intelligent AI-Based Smart Crop Disease Diagnosis and Yield Prediction System.

11. References

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